Lecture 23

First steps in detecting short genomic variation

So what is a variant? Fuzzy Terminology

- **SNPs**, single nucleotide polymorphisms. The difference consists of a single base change.
- **INDEL**s, insertions and deletions. The difference consists of having a single base added or removed.
- **SNV**s, single nucleotide variants. A combination of SNPs and INDELs. The difference is a single base.
- **Short** variations. Variations typically less thanÂ 50bpÂ in length.
- **Large-scale** variations. Variations largerÂ 50bp
- Genomic **rearrangements**. Typically variations on the kilo-base scale.

What about in between?

Who decides what is short, long?

What if something is right at the cutoff of short and long?

Technology vs Reality

The technological limitations determine what variations can be detected with it. Everything else is **lost** from sight!

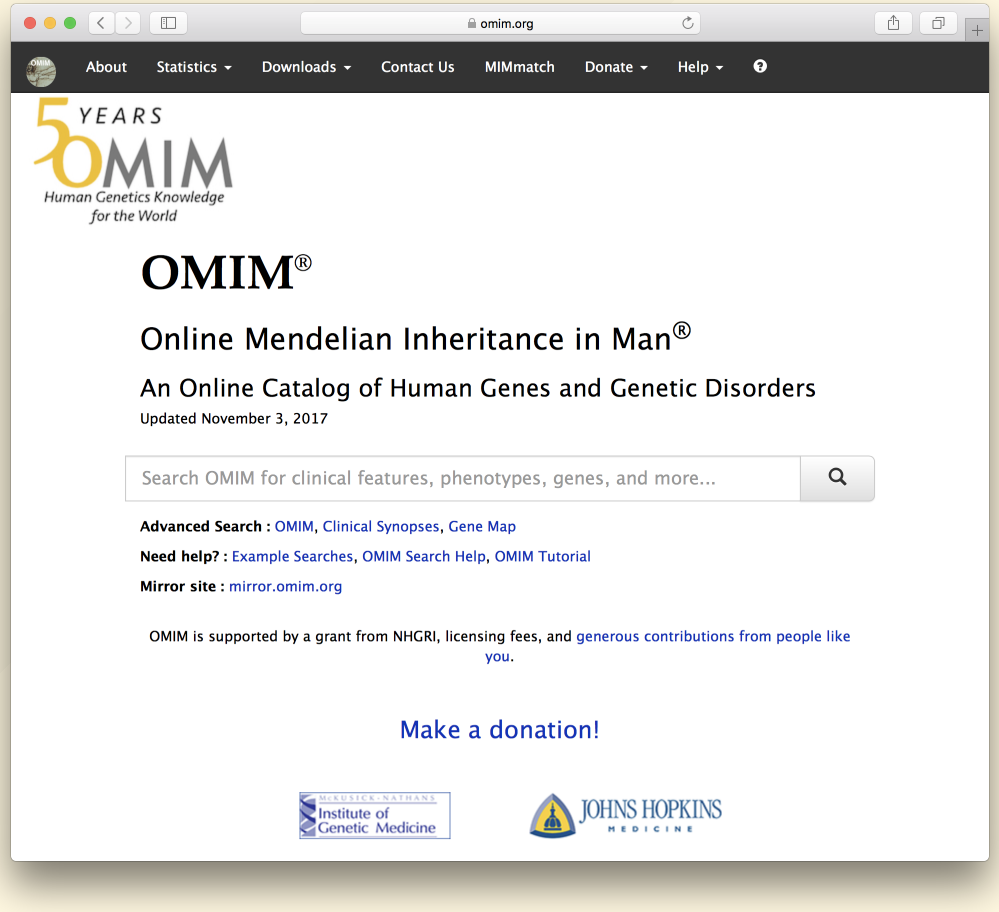
The **streetlight effect** dominates. People look where there is light because it is lighted up there.

We are early in the process understanding what we can reliably detect.

Great news: there is **plenty left to find**! You can use your eyes and mind to find what everyone else missed!

What can be directly explained by variation?

How many phenotypes with known genetic explanations? **Online Mendelian Inheritance in Man**



What is a SNP?

- **SNP**, pronounced snip; plural snips

A DNA sequence variation that **occurs commonly** within a population (e.g. 1%) where a single nucleotide: A, T, C or G of a shared sequence differs between members of a biological species or paired chromosome.

Biologists **love** SNPs because they promise an **easy** explanation to whatever they are looking for:
Well, it's a SNP! Publish it! Case closed. Next!

What is ploidy?

The number of sets of chromosomes in a cell.

The number of possible alleles for a sequence.

Common terminology:

Haploid (1), diploid (2) and polyploid (3 or more).

Reality is more complicated (as always). Not all chromosomes must have the same number of copies. Example human sex chromosomes (X and Y).

More terminology: for diploid organisms

- **Homozygous**: two identical alleles at a given locus.
- **Heterozygous**: two different alleles at a locus.
- **Hemizygous**: only a single copy of a particular in an otherwise diploid organism.
- **Nullizygous**: both copies are missing in an otherwise diploid organism.

Note how complexity manifests itself in everything.

Viewing variations. A graphical pileup



What is a pileup?

Samtools multi-sample pileup: all bases at an

```
samtools mpileup
```

Needs a:

- A SAM/BAM alignment file.
- A genome reference file (optional).

Can produce pileups as well as genotype likelihoods.

How likely is that a given index is covered by an A,T,G,C, deletion or insertion)

What does a pileup look like?

Create a pileup:

```
samtools mpileup -f $REF demo.bam  | head
```

Prints:

```
AF086833.2 46 G 2  ^].^].       @b
AF086833.2 47 A 3  ..^],        <]B
AF086833.2 48 A 3  ..,         @cE
AF086833.2 49 T 3  ..,         FgH
AF086833.2 50 A 3  ..,          FdF
AF086833.2 51 A 5  ..,^].^]. Fg@C@
AF086833.2 52 C 7  ..,..^].^]. DgGCBCC
AF086833.2 53 T 7  ..,.... DgHC@CC
AF086833.2 54 A 7  ..,.... HiEFFCC
AF086833.2 55 T 7  ..,.... HkHFFFF
```

Weird symbols that explain what happens at each coordinate

Why all those details?

Understanding alignments is key to understanding variants.

You want to make sure the alignment is correct - or that the call is right.

- `.` a match on the forward strand
- `,` a match on the reverse strand
- lower case letter, a mismatch on the forward strand
- capital letter, a mismatch of the reverse strand

Pileup examples. What does this mean?

```
G..,....gg,,...GG,,.gg
```

- 10 bases match on the forward strand,
- 5 bases match on the reverse strand
- 4 bases indicate a mismatch of `g`
- 3 bases indicate a mismatch of `G`

15 bases indicate no mismatch, 7 bases indicate a mismatch and all agree on the mismtch (G).

Would you trust this variant? Is this homozygous, heterozygous? Something else?

Starts/ ends of alignments are more suspicious

Sequences misalign more frequently at starts and ends.

Special characters indicate bases that align at start and end.

- `^` and `$`

See the book course page for links to full description.

A variant caller reads off all these signals and tries to reconcile them.

What are phased variants?

When there is more than one copy knowing which variants are on the same DNA is called "phasing."

Two variants are in phase if they form the same haplotype (inherited together).

"Inherited together" is misleading since they aren't always inherited together (chromosomal recombination can break that). Most often though they do.

Unphased variants are genotypes where we don't know which chromosomes hold that allele.

"Naive" Variant calling

Find short variants directly from the pileup

Works very well if the alignments are "correct" and the genome is "densely packed" with information - few redundant segments.

Bacteria, viruses in general (but not always).

It becomes an issue of statistics - how many variants would you need to see to trust it.

It is also a matter of population: clonal or not. How many alleles could there be?

If you only need to check a few variants

You can visually look them up and figure out reasonably well yourself what they are.

Especially true if your problem is subtle, and has unexpected qualities to it.

Your eye will win out over any variant caller - if you are even a little bit trained in what to do.

What does empirical variant calling look like

Example: We found a previously unknown isoform of a transcript that contained a single edited base relative to the standard transcript.

We did it all by eye.

The biologists expected it to see it at a location, sure enough, it was there but shallow coverage. The transcript is very rare. And we found it over and over again in other scientists' samples.

It is always there, in tiny quantities, always the same way. No variant caller would ever find it.

Real variant calling

What if the alignments are "incorrect"?

Incorrect does not mean the mathematics of alignment is wrong!

Alignments find the simplest and most efficient explanation based on a scoring matrix.

But what if you can't possibly know which scoring matrix is the right one?

Advanced variant callers can produce **right calls** from **"wrong" alignments**! It is amazing!

Important take-home lesson

You too can be a **Variant Caller**!

We use software baseed variant callers because we can't check everything ourselves.

There are just too many variants to look at.

If you know what you are looking for, you can do a better job with pileups and by eye.

A variant caller weighs data by statistical measures that cannot incorporate your knowledge and expectations.